SEMESTER 7 · COMPUTER VISION PROJECT

CASE FILE
NIH · 27,558

NIH · THIN SMEAR · GIEMSA · P. FALCIPARUM

Automated *Microscopic* Diagnosis of Malaria

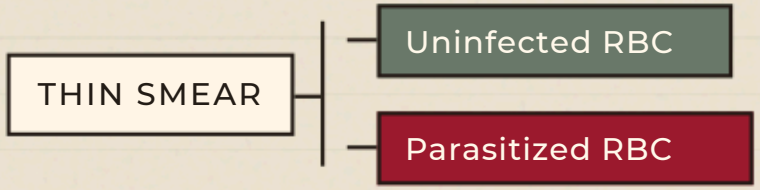
NIH cell patches · Giemsa thin smears · Chittagong Medical College Hospital

Made by — Agrapujya Lashkari, Deepanshu Mehta

LITERATURE SURVEY · 110 PAPERS · 2018–2025

THE PROBLEM

A person still sits at the microscope.



27,558

CELL PATCHES

13,779

EACH CLASS · BALANCED

200 pts

150 INFECTED · 50 HEALTHY

Chittagong

RELEASED BY NIH

NIH · THIN SMEAR · GIEMSA · P. FALCIPARUM

Literature findings

01 Rajaraman et al., 2018 · *PeerJ* · transferCNNs

02 Rajaraman et al., 2019 · *PeerJ* · ensemble

03 Rahman et al., 2019 · arXiv · custom CNN

04 Fuhad et al., 2020 · *Diagnostics*

05 Masud et al., 2020 · mobile CNN

06 Rajaraman et al., 2018 · MIUA · XAI

07 Madhu et al., 2023 · CapsNet · *Sci. Rep.*

08 Masud et al., 2022 · CCT transformer

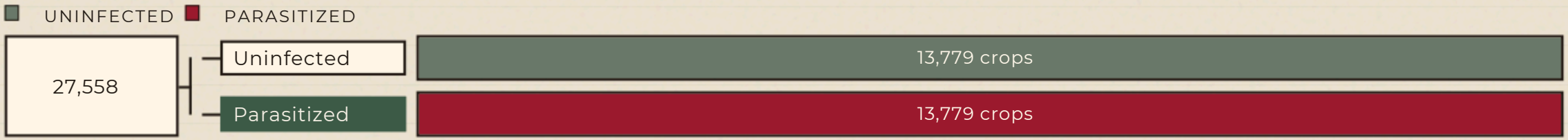
09 Zedda et al., 2023 · YOLO-PAM

10 Mmileng et al., 2025 · ConvNeXt V2 Tiny

11 Ali et al., 2025 · DANet · *Sci. Rep.*

Rajaraman et al. — transfer CNNs

RESNET-50 / IMAGENET | FEATURE EXTRACTORS | NIH CELLS



PROBLEM
Not enough labeled smears to train a deep net from scratch. Manual features were brittle.

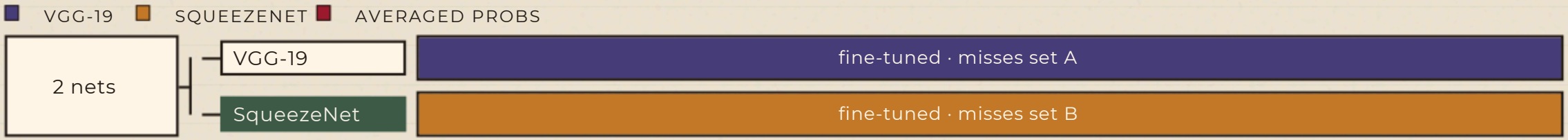
HOW IT WAS SOLVED
ImageNet CNNs (ResNet-50 and kin) as frozen / transferred feature extractors on cell patches.

LIMIT
A single model jumped with small training-set quirks. High prediction variance.

02 / 101 · PEERJ 2019

Rajaraman et al. — ensembles

VGG-19 + SQUEEZENET | MODEL AVERAGING | COMPUTE-HEAVY



PROBLEM
One transfer model is jumpy. Unreliable for a diagnostic call.

HOW IT WAS SOLVED
Average output probabilities of two nets that fail in different places.

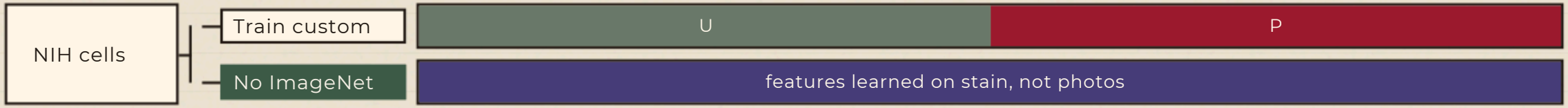
LIMIT
Too heavy for low-power clinics where point-of-care malaria testing actually happens.

03 / 101 · ARXIV 2019

Rahman et al. — custom CNN

END-TO-END CNN | FROM SCRATCH | RAW RBC PATCHES

U P · LEANER THAN THE ENSEMBLE



PROBLEM
Ensembles were too bulky for field deployment.

HOW IT WAS SOLVED
A purpose-built CNN on raw red-cell patches. Skip the ImageNet stack.

LIMIT
Still a black box. Unclear if the net sees Plasmodium or stain debris.

Fuhad · Masud — phone-scale nets

ULTRA-LIGHT CNN | SMARTPHONE / IOT | BLACK BOX

FUHAD ET AL., 2020

Deep learning on blood smears with a smartphone application. Designed for the device in the clinic, not a GPU rack.

MASUD ET AL., 2020

Mobile malaria detector. Same goal: inference that fits cheap hardware after cleaning mislabeled anomalies.

PROBLEM

Point-of-care sites cannot run VGG ensembles.

HOW IT WAS SOLVED

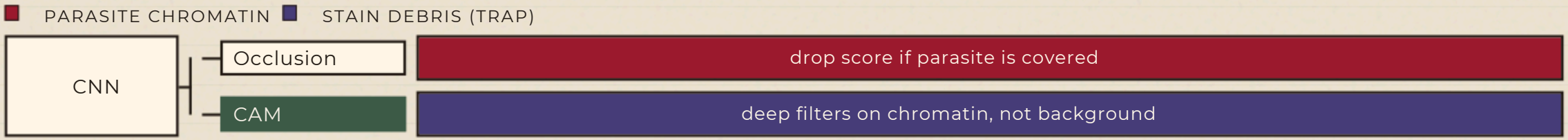
Tiny CNNs on cell patches. Scrub obvious label noise so the small net still trains.

LIMIT

Interpretability is gone. A doctor cannot tell parasite from dye speck.

Rajaraman et al. — XAI

OCCLUSION MAPS | CAM | CLINICAL TRUST



PROBLEM
Clinicians distrust a lightweight CNN they cannot insp e ct.

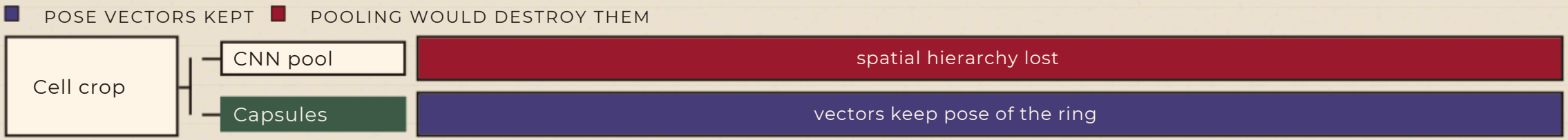
HOW IT WAS SOLVED
Occlusion sensitivity and class activation maps show the filters tracking the parasite.

LIMIT
Max-pooling still shreds pose, shape, and orientation of the parasite.

NIH · THIN SMEAR · GIEMSA · P. FALCIPARUM

Madhu et al. — CapsNet

INCEPTION CAPSNET | DYNAMIC ROUTING | 99.25% ACC



PROBLEM
Pooling destroys the parasite’s geometry.

HOW IT WAS SOLVED
Inception-based capsules with dynamic routing. Output vectors hold spatial pose.

LIMIT
99.25% on pre-cropped cells. Math is heavy. Edge deployability undone.

08 / 101 · SENSORS 2022

Masud et al. — compact transformer

MULTI-HEAD ATTENTION | CCT | GLOBAL CELL CONTEXT

PROBLEM
CNNs see local patches. Ring-stage parasites need the whole cell as context.

HOW IT WAS SOLVED
Compact Convolutional Transformer. Self-attention over the entire cell at once.

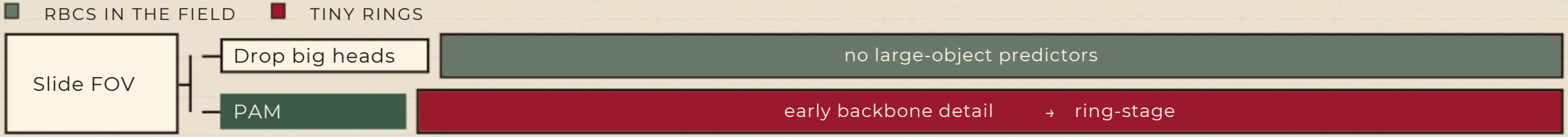
LIMIT
Same ceiling as CapsNet: high accuracy, high cost, still one cropped cell at a time.

> CapsNets and transformers pushed cell-level scores into the high nineties. They also walked the field back from phones toward GPUs.

09 / 101 · J. IMAGING 2023

Zedda et al. — YOLO-PAM

YOLOV8 | PARASITE ATTENTION | WHOLE FIELD OF VIEW



PROBLEM

Clinics do not crop single cells. The slide is overlapping and messy.

HOW IT WAS SOLVED

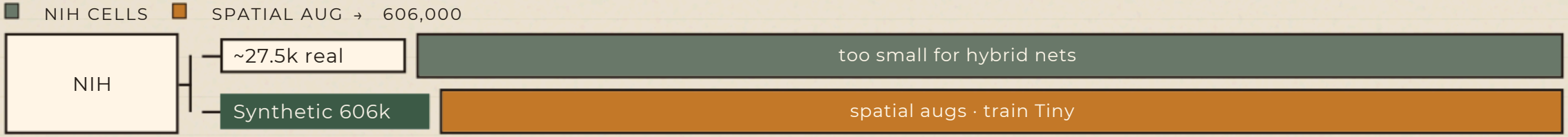
YOLOv8 minus large-object heads, plus a Parasite Attention Module on high-resolution features.

LIMIT

Fragile under data scarcity. Shear augmentations warp tiny rings until the detector misses them.

Mmileng et al. — ConvNeXt + synthetic

CONVNEXT V2 TINY | 606K SYNTHETIC | ADAMW · LABEL SMOOTHING



PROBLEM

Advanced models want huge data. Clinics want tiny compute.

HOW IT WAS SOLVED

ConvNeXt V2 Tiny (ViT hierarchy, CNN speed) on 606k synthetic images. Label smoothing + AdamW.

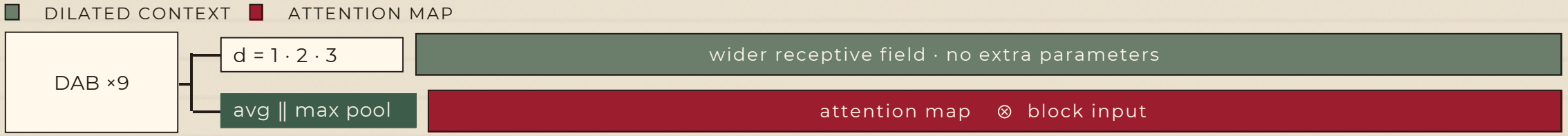
LIMIT

Fake data + pretrained weights. Risk of domain shift on unseen stain or strain.

11 / 11 · SCI. REPORTS 2025

Ali et al. — DANet

DILATED ATTENTION | 2.3 M PARAMS | RASPBERRY PI 4



PROBLEM

Hybrid CNN–transformers win accuracy, not clinics. Low-contrast smears blur the parasite’s b orders.

HOW IT WAS SOLVED

Nine dilated attention blocks (d = 1, 2, 3) widen context for free. 97.95 % accuracy with 2.3 M p arams.

LIMIT

Random cell-level split of one malaria dataset. Grad-CAM shown, not validated. No regularization.

DO NOT RANK ON ONE BOARD

Same NIH cells. Different exams.

#	PAPER	MOVE	UNIT	HEADLINE	COST OF THE WIN
1	Rajaraman ’18	Transfer CNN	cell	ImageNet works	high variance
2	Rajaraman ’19	Ensemble	cell	stable probs	too heavy for clinic
3	Rahman ’19	Scratch CNN	cell	end-to-end	black box
4–	Fuhad / Masud	Tiny CNN	cell	phone / IoT	no explanation
5 6	Rajaraman XAI	CAM / occlusion	cell	looks at parasite	pooling kills pose
7	Madhu CapsNet	capsules	cell	99.25%	not edge-ready
8	Masud CCT	transformer	cell	global context	still one crop
9	YOLO-PAM	detection	FOV	tiny rings	aug / data fragile
10	ConvNeXt Tiny	hybrid + synth	cell	606k images	**domain shift**
11	DANet (Ali ’25)	dilated attention	cell	2.3 M params	cell split only

PHASE 1 · STILL OPEN

Four gaps before a clinic.

01 · CELL-LEVEL SPLIT LEAK

Random cellsplitslet the net memorizeapatient’sstain.Group by patient, or the nineties are an illusion.

02 · DOMAIN GENERALIZATION

NIH or synthetic clones fail on a smear from another region, dye, or strain.

03 · INSTANCE-LEVEL DETECTION

Real slides are thick and thin fields, not pre-cropped cells. Multi-instance, real-time detection is the clinical unit.

04 · CLINICALLY VALIDATED XAI

Grad-CAM is not enough. Attention must match parasitological markers: hemozoin, chromatin dots.

CASE FILE
M2 · M3 · DANET

PHASE 2 · MILESTONES 2 – 3

Reproduce DANet.

Ali et al. · Scientific Reports 2025 · a 2.3 M-parameter dilated attention network

Made by — Agrapujya Lashkari, Deepanshu Mehta

BASELINE · REPRODUCTION GAP · HYPOTHESIS · PREDICTION BEFORE CODE

M2 · BASE PAPER · SCI. REPORTS 2025

DANet — attention on a 2.3 M budget.

9 DILATED ATTENTION BLOCKS | 2.31 M PARAMS | 0.83 GFLOPS / CELL | CODE ON GITHUB | PAPER: 97.95 % ACC

ONE DILATED ATTENTION BLOCK · AS RELEASED IN CODE



CODE AS SHIPPED

Crashes on import (NameError). The output layer has 32 classes for a 2-class task. Two small fixes.

PAPER≠ CODE

Text: parallel dilations, sigmoid gate, LayerNorm. Code: cascaded, softmax, InstanceNorm. We ran the code: 2.31 M params matches.

OPTIMISER TRAP

Table 2's lr 0.01 never leaves 50 %. A default OneCycle call silently adds momentum. It trains with the text's 0.001, momentum 0.

M2 · REPRODUCTION · 75 EPOCHS · SEED 42 · 5,512 TEST CELLS

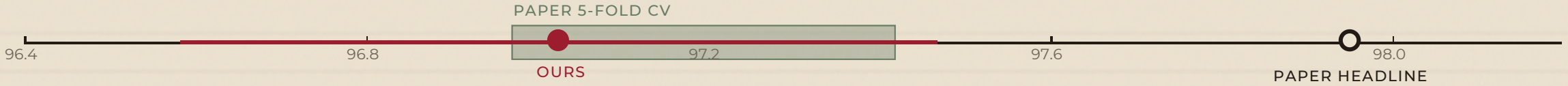
97.0 %. Inside the paper's *own* CV range.

97.02%
ACCURACY · PAPER 97.95

96.30%
SENSITIVITY · PAPER 98.07

97.75%
SPECIFICITY · PAPER 97.87

0.9945
ROC-AUC · PAPER 0.98



Accuracy, %. The red bar is our 95 % CI. The paper's headline 97.95 beats all five of its own CV folds. Our gap is almost all **sensitivity**: 102 missed infections vs 62 false alarms.

M3 · HYPOTHESIS · ONE VARIABLE

Hide the stain. Make it read the *parasite*.

If we add **stain-colour augmentation** to DANet's training, **sensitivity on unseen patients will rise**, because each slide's stain is a fingerprint the network can use in place of parasite shape and chromatin.

ONE CHANGE · COLOUR JITTER | SATURATION ±50 % · HUE ±0.1 | BRIGHTNESS / CONTRAST ±10 % | ALL ELSE FROZEN

MEAN CELL COLOUR · 201 PATIENTS · ONE SWATCH EACH · SORTED BY HUE



MECHANISM

Random colour on every image breaks the stain → patient → label shortcut. Only morphology stays reliable.

EVIDENCE

Patient ID explains 74–90 % of colour. The 50 healthy-only patients supply 25 % of the uninfected cells.

WHY WE MIGHT BE WRONG

The InstanceNorm in every block already strips global colour, and mean colour alone predicts the label at only 61 %.

M3 · HYPOTHESIS · ONE VARIABLE

Architectural De-Coupling (Parallel vs. Cascaded Dilations)

Modify the Dilated Attention Block (DAB) to use parallel dilated convolutions (as described in the paper's text) rather than the cascaded dilations found in the released code.

The Prediction: Sensitivity will slightly decrease (or remain stagnant), but the parameter count will drop.